LEARNER GUIDE

The angle bisector

Constructions, Loci and Bearings · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Two straight roads meet at a junction, and you want a path that stays the **same distance from both** as it heads away. There is exactly one such path — it runs up the middle of the angle, and it is the **angle bisector**. Every point on it is equidistant from the two **arms** of the angle. This lesson rhymes with the last: there you found the points equidistant from two *points* (the perpendicular bisector); here you find the points equidistant from two *lines*, and construct the bisector exactly with a compass.

What you are learning to notice

- A point the same distance from both arms of an angle is **equidistant** from them.
- **All** the equidistant points form a straight line from the vertex — the **angle bisector**.
- The bisector splits the angle into **two equal halves**.
- It is the **fold line** that lays one arm exactly onto the other — that symmetry is why the halves are equal.

Moving through it, one step at a time

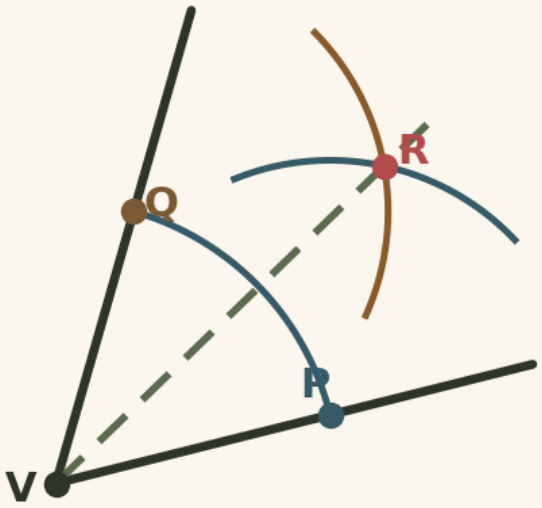
To construct it: put the compass point on the vertex and swing an arc that cuts both arms, at P and at Q. Then, keeping one opening, swing an equal arc from P and from Q so they cross inside the angle at R. Draw the straight line from the vertex through R: that is the angle bisector. Because both arcs were equal, the figure is symmetrical about this line, so it halves the angle. Leave your arcs showing.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Construct the bisector of an angle

Construct the bisector of the angle at V.

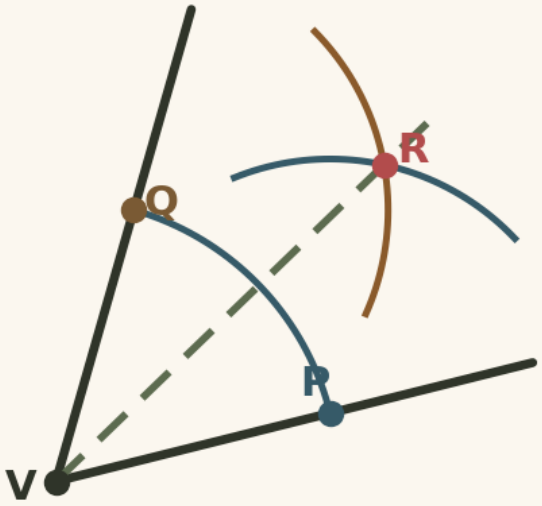


1. From V, swing an arc cutting both arms at P and Q.	arc from V
2. Swing an equal arc from P.	arc from P
3. Keep the opening; swing an equal arc from Q, crossing at R.	arc from Q
4. Draw the line from V through R.	join V–R

V–R is the angle bisector: it splits the angle into two equal halves, and every point on it is the same distance from both arms.

Example 2 — Halving an angle

The bisector splits the angle into two equal halves. If the whole angle is 80°, how big is each half?



1. Construct the bisector V–R (as above).	line V–R
2. It divides the 80° angle into two equal parts.	80 ÷ 2
3. Each half is 40°.	= 40°

Each half is 40° — the bisector halves the angle exactly, with no protractor.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (equal-distance compass arcs (Lesson 1) and the perpendicular bisector (Lesson 3)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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